

Astronomy Program Learning Outcomes (new version, letter of intent submitted early 2026, must track progress)						
1 Demonstrate knowledge of factual material essential to the physical sciences. 2 Apply the scientific process, including conducting experiments, testing hypotheses and interpreting results. 3 Prepare written reports in a standard scientific format. 4 Analyze and interpret quantitative scientific data. 5 Perform laboratory skills in the physical sciences. 6 Evaluate information using scientific knowledge and reasoning.						
Plan year	PLO(s) to be assessed	Course	Tool(s) being used	Desired performance	Timeline	Lead person responsible for data collection, analysis, and reporting.
Year 1(2026-27)	PLO 1	AST101 and AST102	Test of Astronomy Standards (TOAST)	Average scores equal or better than national average	Completed by end of Spring 2027	Glenda Denicolo, Matthew Pappas
Year 2(2027-28)	PLOs 2, 3, 4, 5	PHY 132	Laboratory report with standardized rubric.	2/3 of our students will meet or exceed standards on each of the topics covered in the scoring rubric.	All PHY132 sections will be assessed during Fall 2027 and Spring 2028.	Scott Gianelli, Darin Mihalik
Year 3(2028-29)	PLO 1	AST101 and AST102	Test of Astronomy Standards (TOAST)	Average scores equal or better than national average	Completed by end of Spring 2029	Matthew Pappas, Darin Mihalik
Year 4(2029-30)	PLOs 2, 3, 4, 5	PHY 232	Laboratory report with standardized rubric.	2/3 of our students will meet or exceed standards on each of the topics covered in the scoring rubric.	All PHY232 sections will be assessed during Fall 2029 and Spring 2030.	Scott Gianelli, Darin Mihalik
Year 5(2030-31)	PLO 6	PHY 222	Oral presentation on topic selected by the student, under the instructor’s supervision.	2/3 of our students will meet or exceed standards on each of the topics covered in the scoring rubric.	All PHY222 sections will be assessed during Fall 2030 and Spring 2031.	Scott Gianelli